

PERMANENT MAXILLARY SECOND PREMOLAR

**Dr Saba Arshad (Senior
Lecturer)**

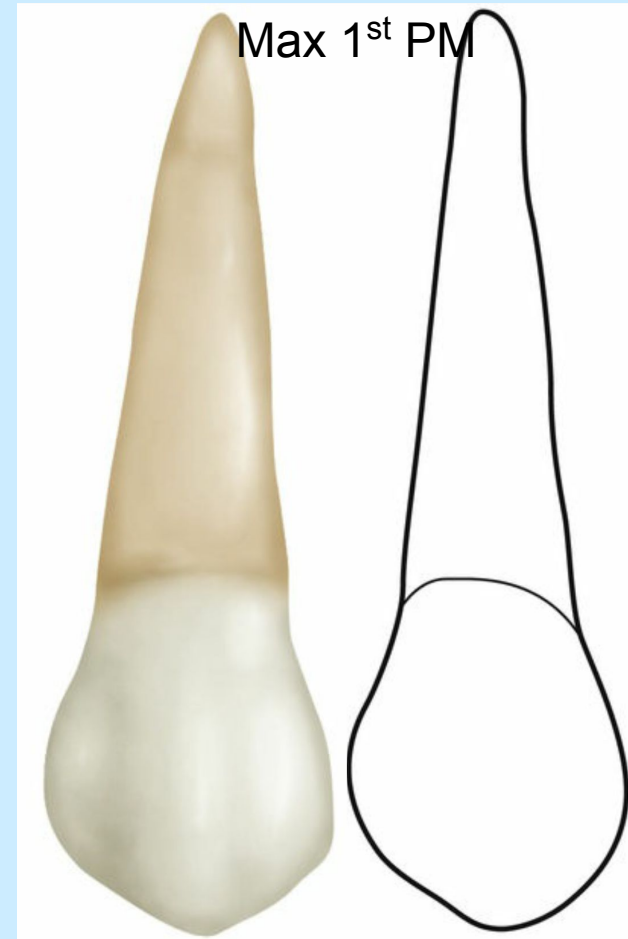
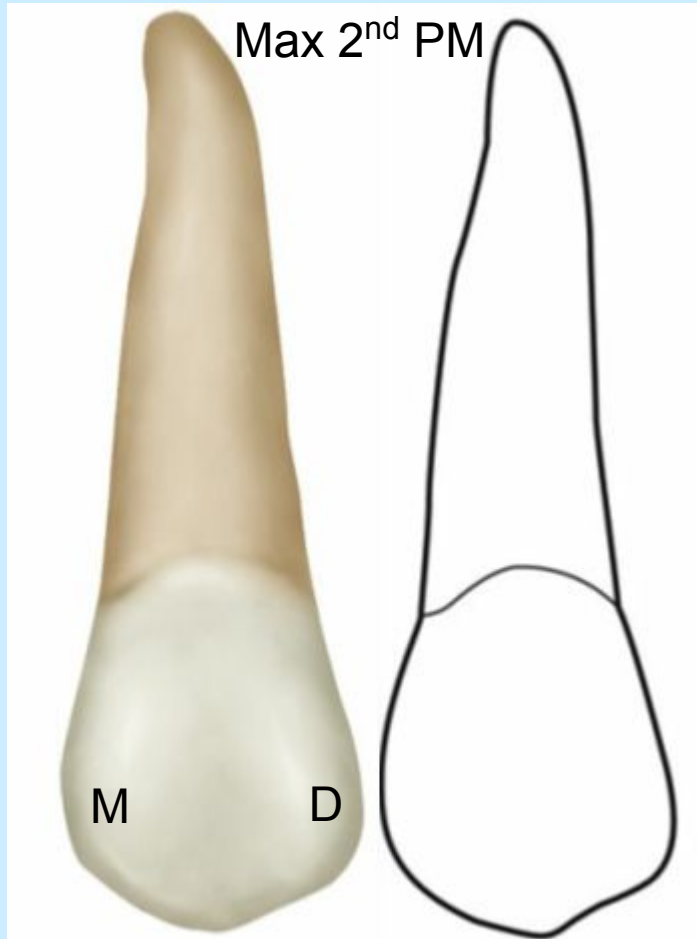
Department of Oral Biology



Learning Objectives

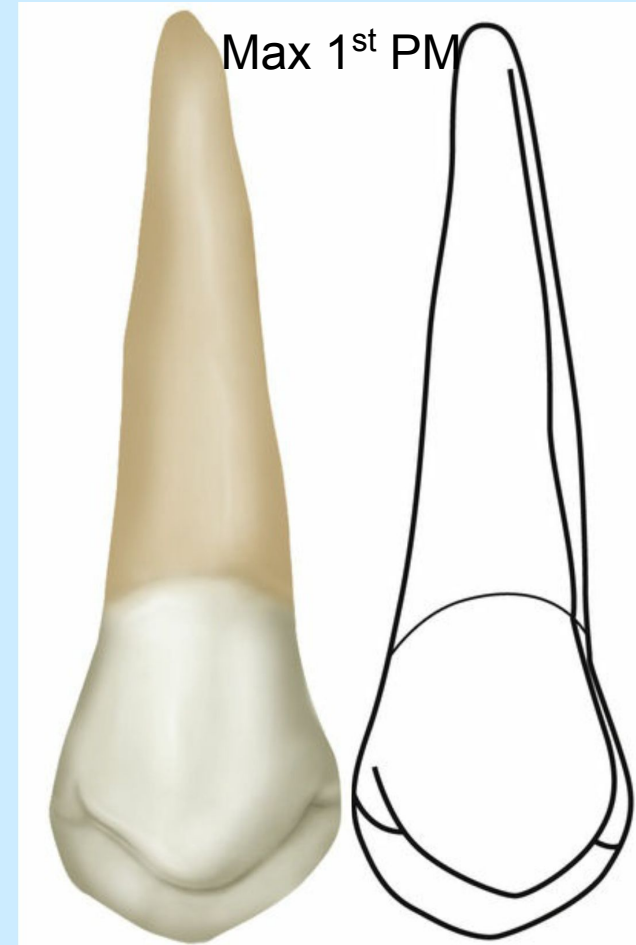
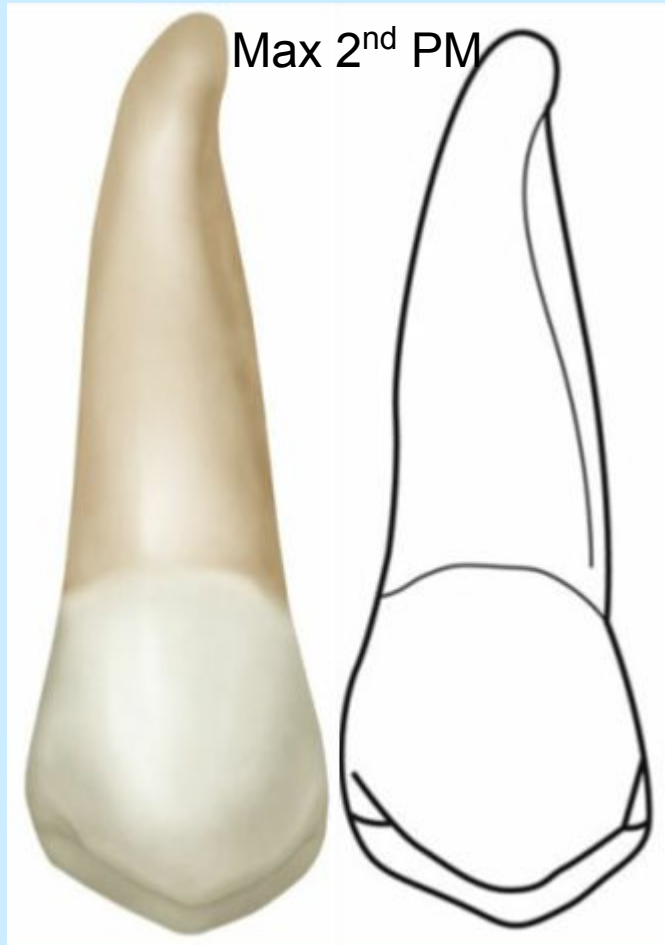
- To define and pronounce the terminologies of permanent posterior dentition.
- To label the anatomical landmarks of premolars.
- To discuss the curves and segments of the teeth.
- To be able to correctly draw the graph outline.
- To be able to carve the teeth on wax models.
- To discuss anatomical variations.

Buccal Aspect



Maxillary left second premolar

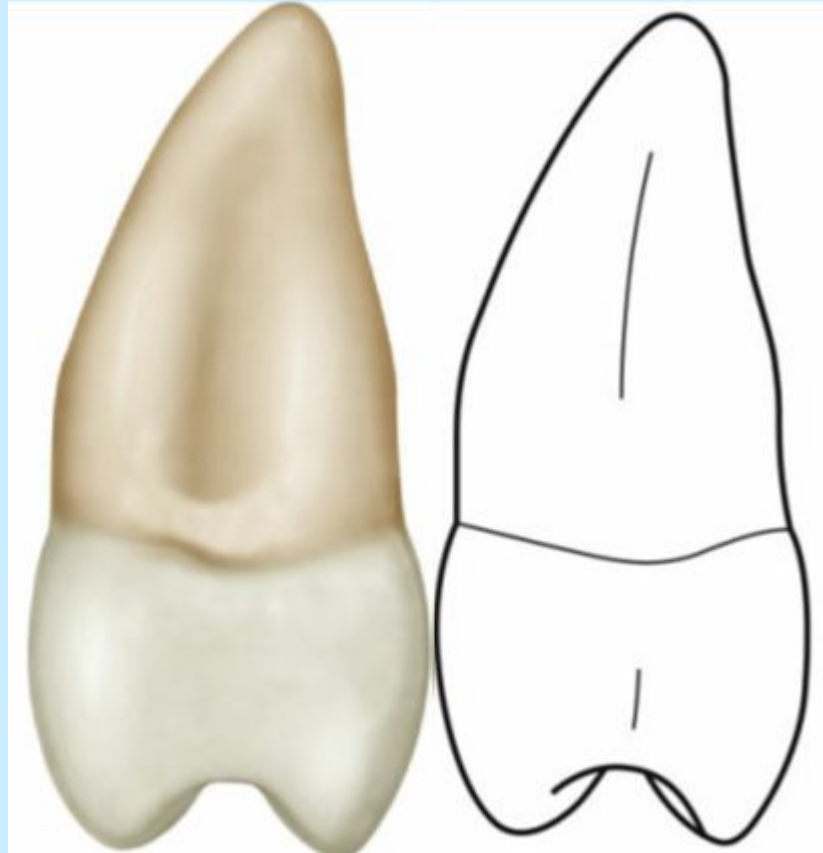
Lingual Aspect



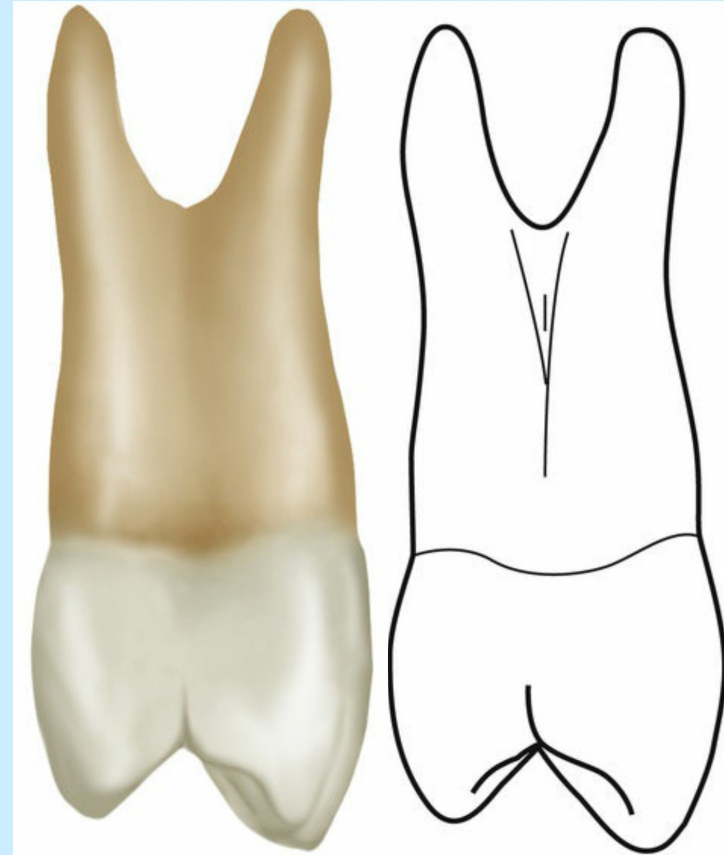
Maxillary left second premolar

Mesial Aspect

Max 2nd PM



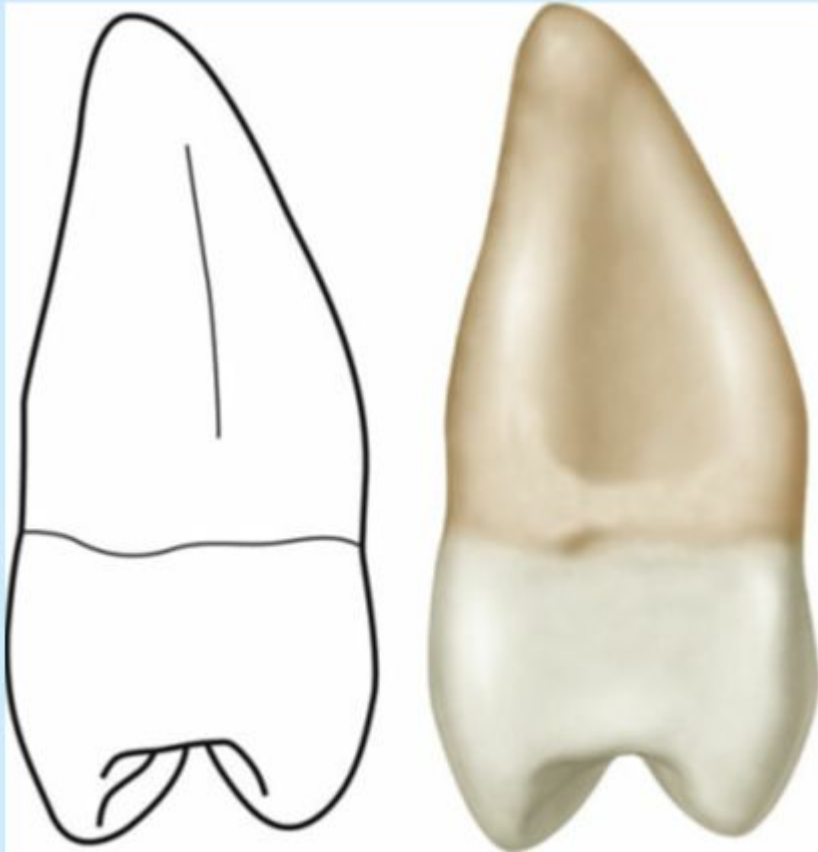
Max 1st PM



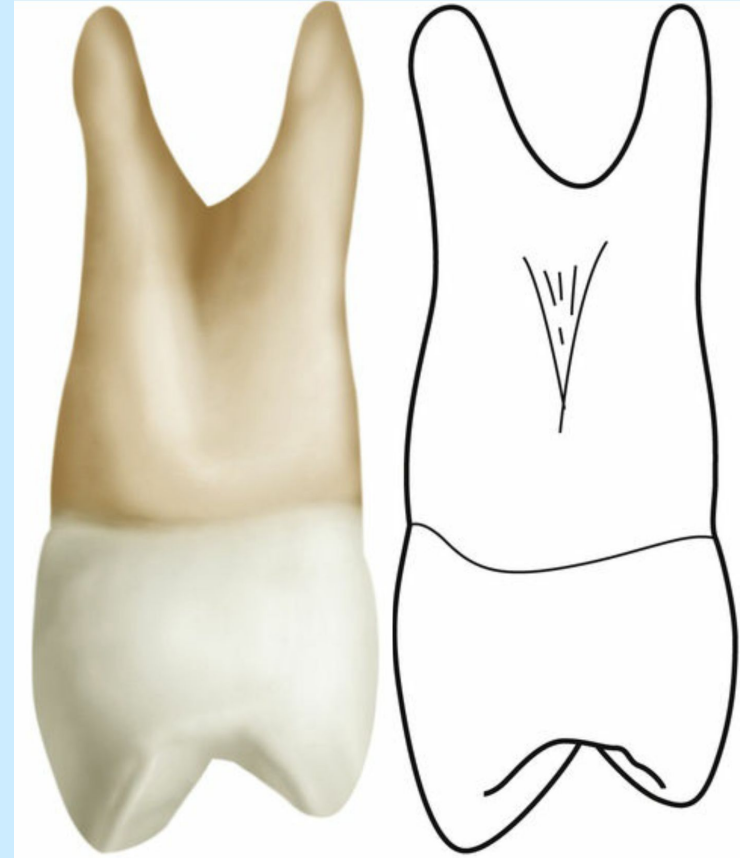
Maxillary left second premolar

Distal Aspect

Max 2nd PM

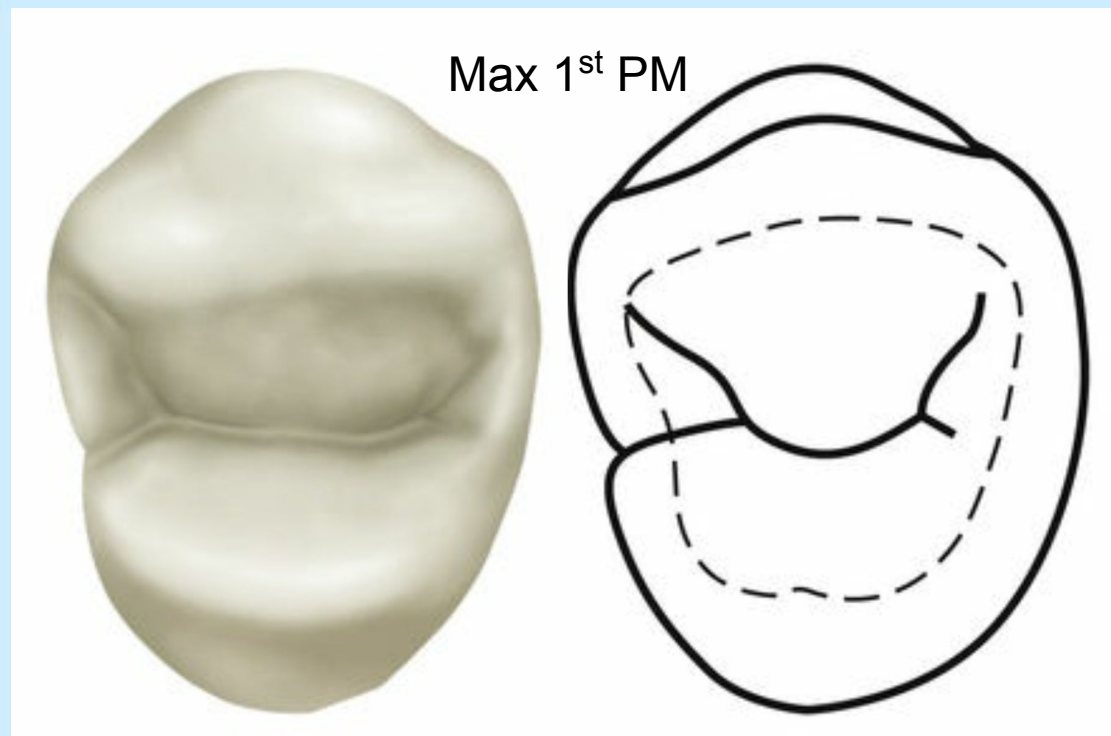
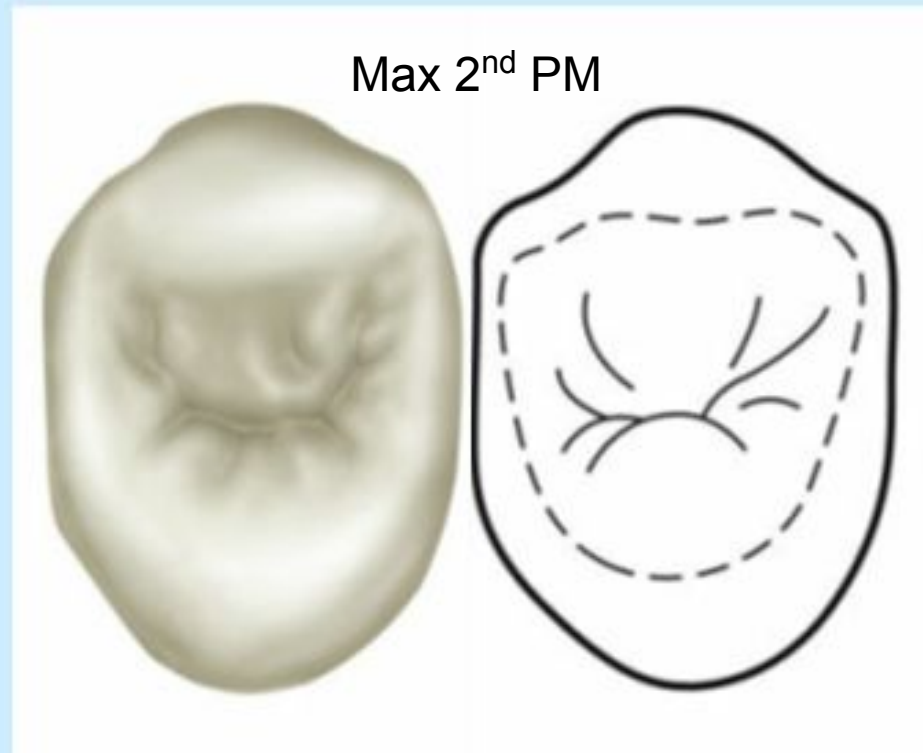


Max 1st PM



Maxillary left second premolar

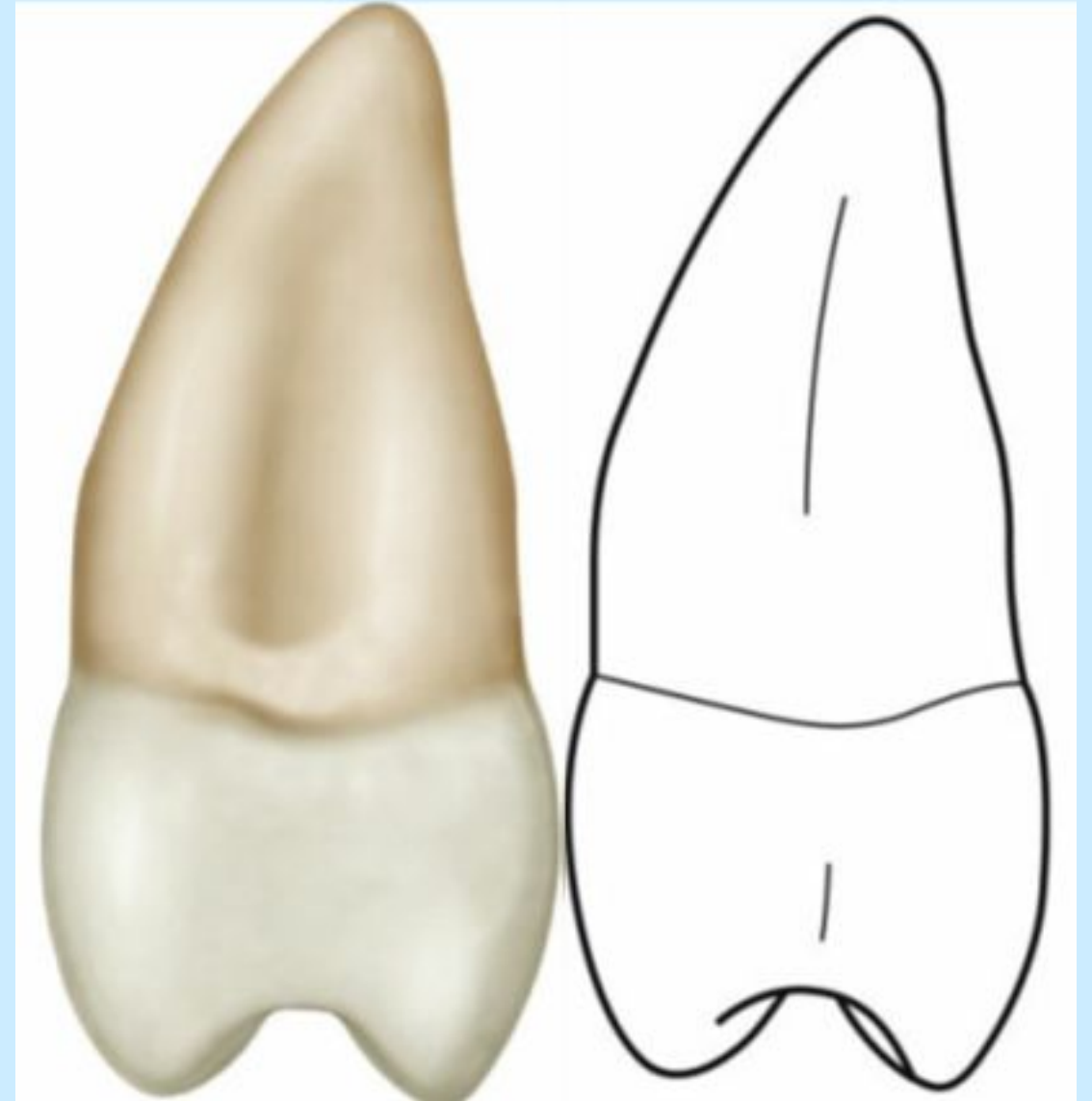
Occlusal Aspect



Maxillary left second premolar

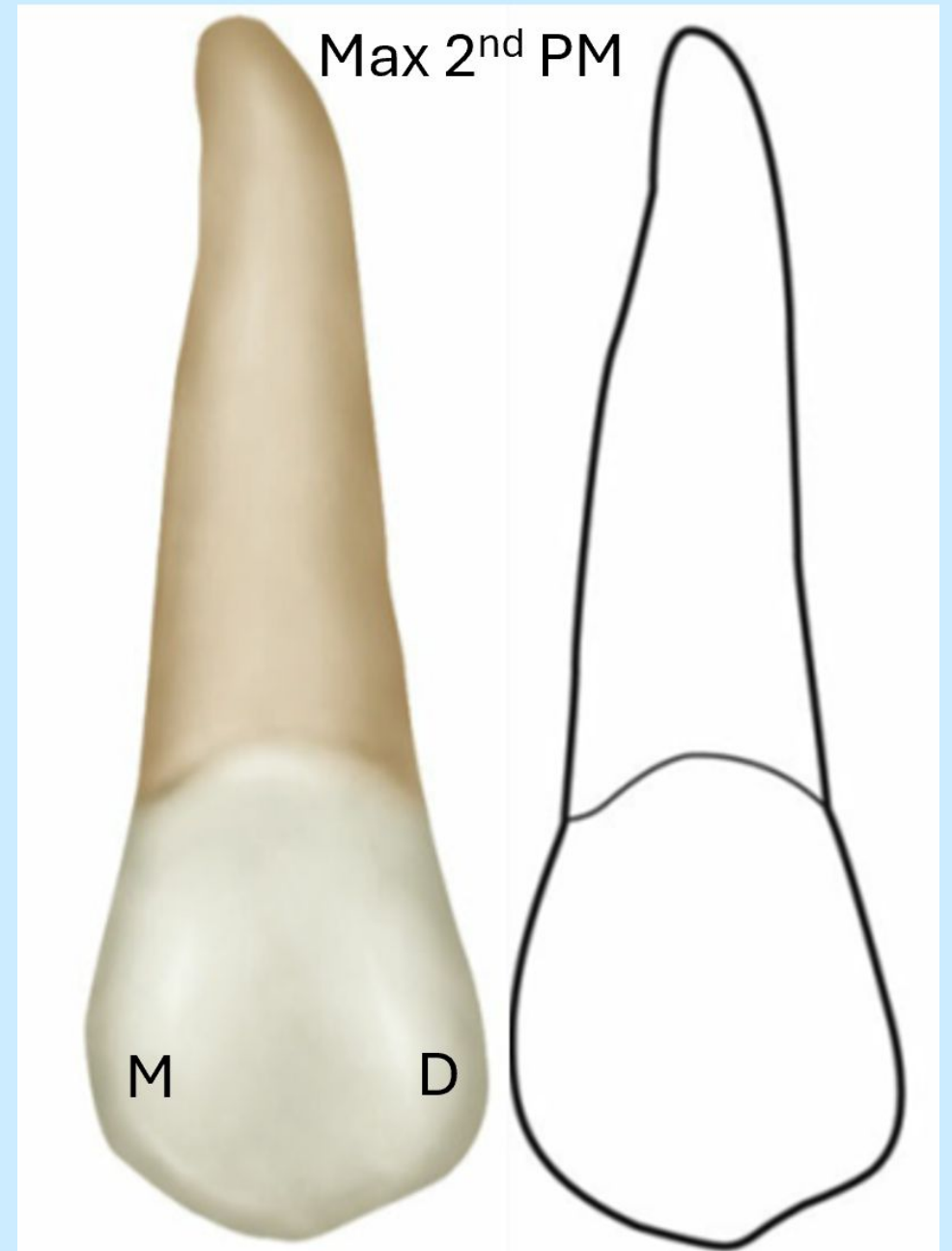


- Two cusps (facial & lingual) of **nearly equal size and height**
- Supplements the 1st premolar in function



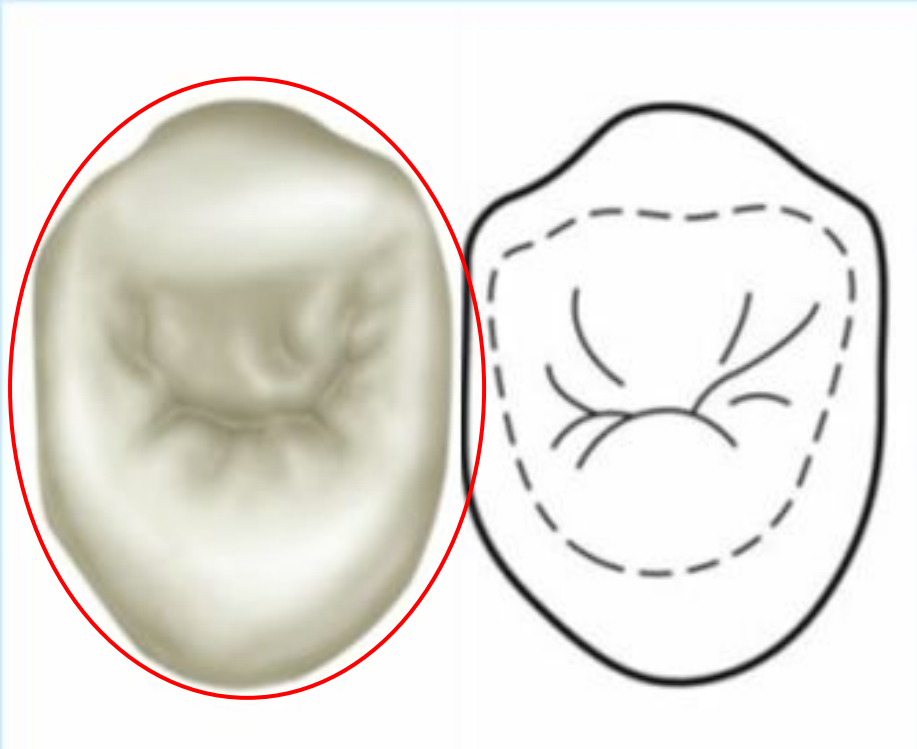
Maxillary left second premolar

- Greater variability than the 1st premolar
- **Distal slope of distal cusp ridge longer than mesial slope of mesial cusp ridge**

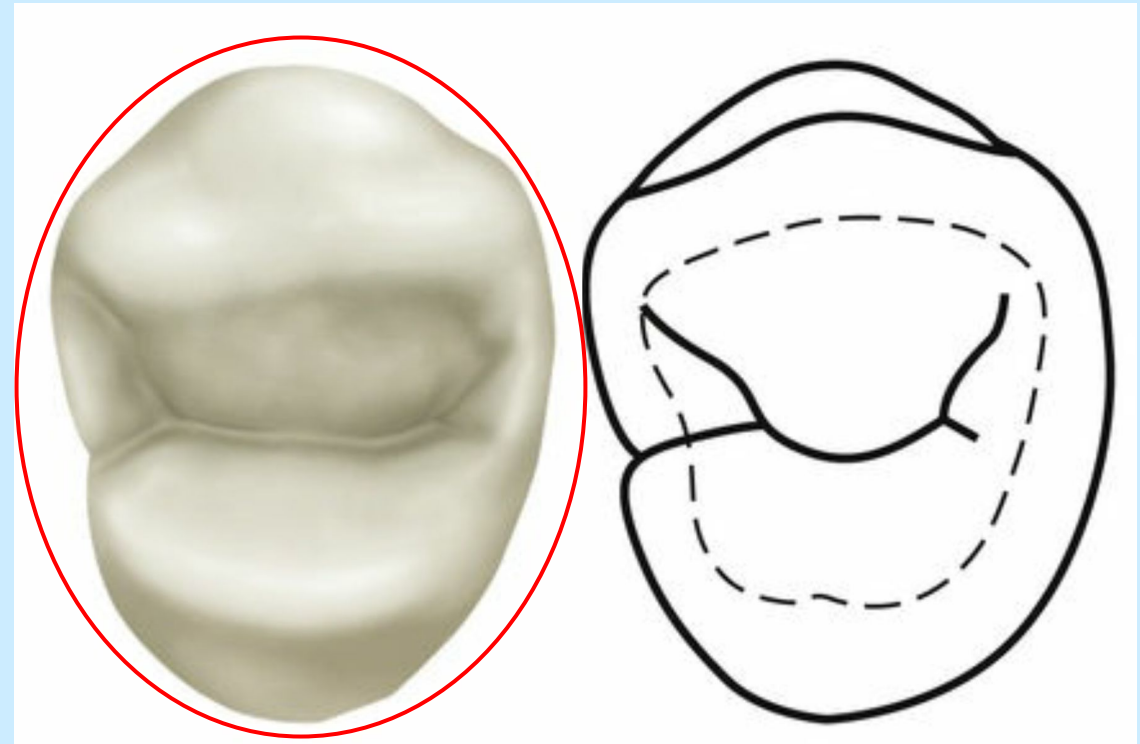


Maxillary left second premolar

- Greater variability than the 1st premolar
- Crown outline and mesiodistal dimensions may be **smaller or larger**



Maxillary left second premolar

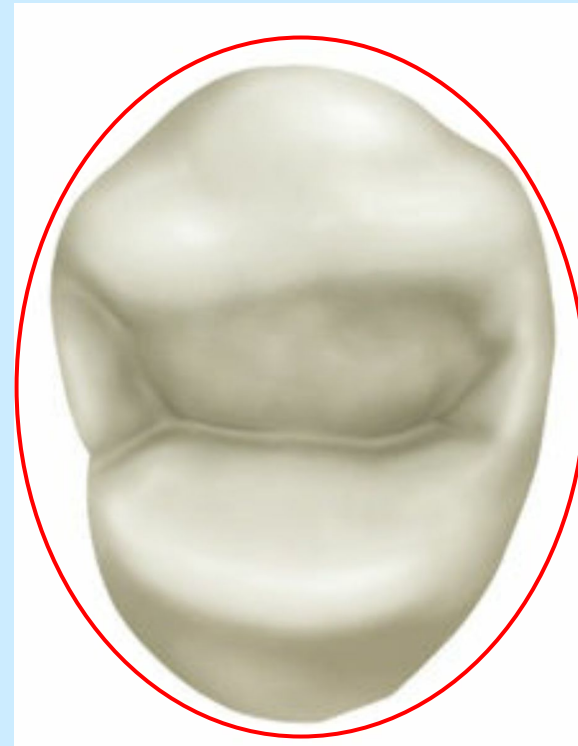


Maxillary left First premolar

- Greater variability than the 1st premolar
- Less angular, more rounded morphology

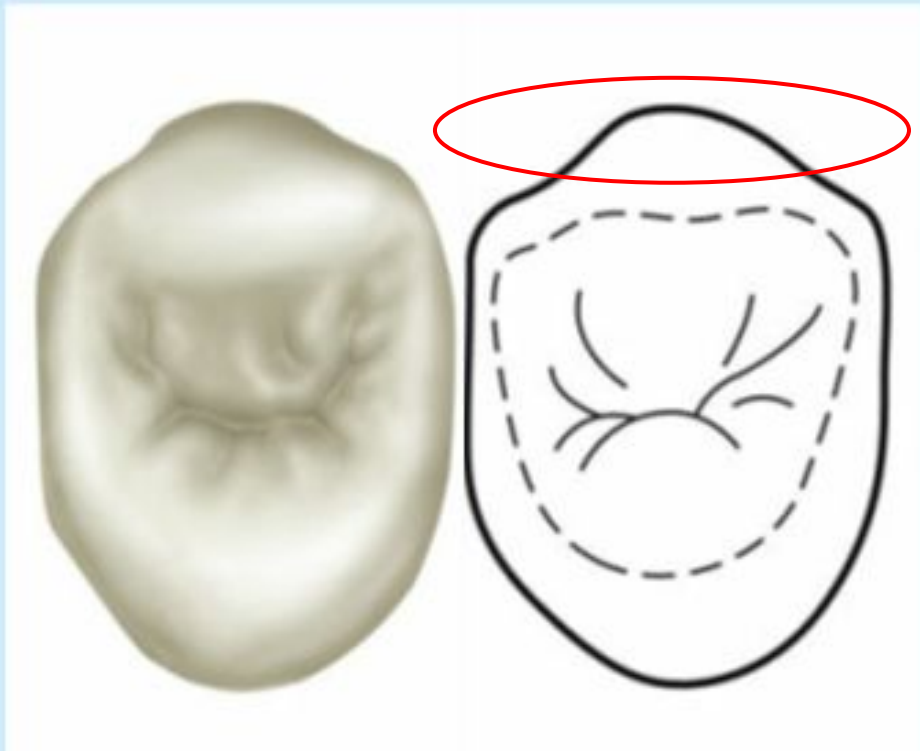


Maxillary left second premolar

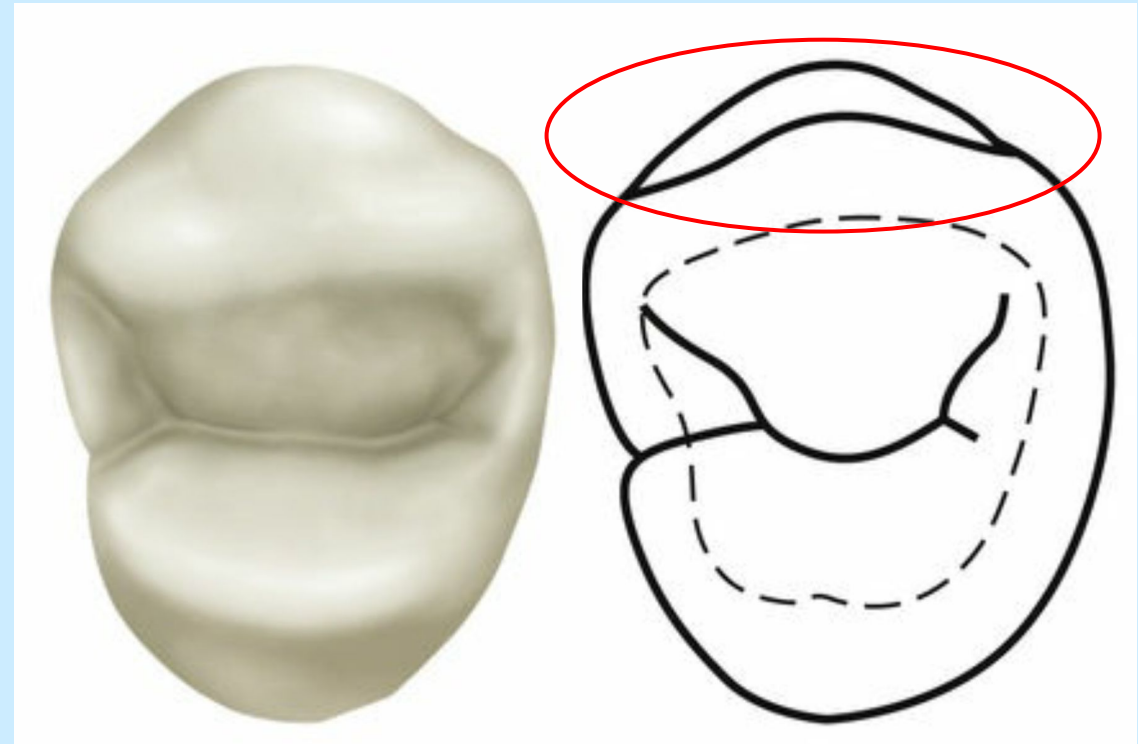


Maxillary left First premolar

- Greater variability than the 1st premolar
- **Facial ridge less prominent than in the 1st premolar**



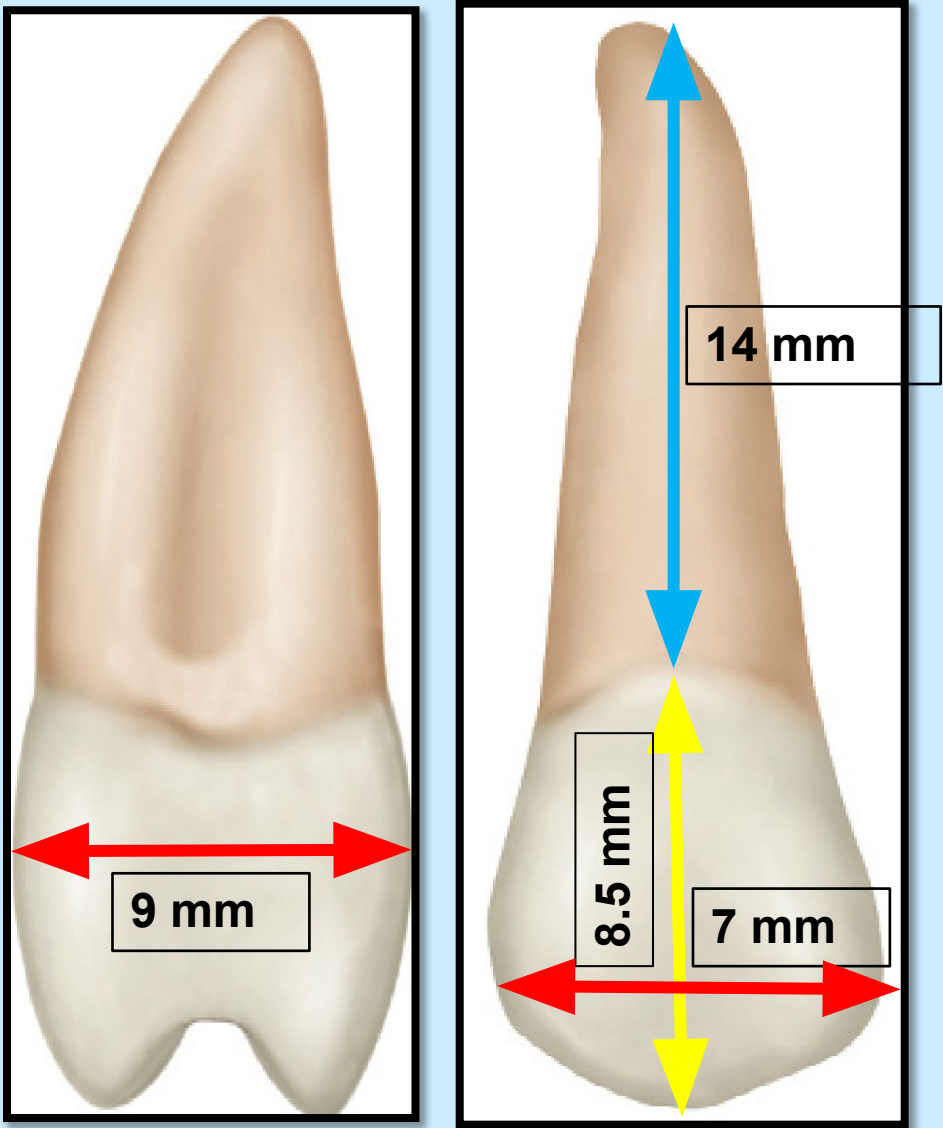
Maxillary left second premolar



Maxillary left First premolar

- Both teeth have similar average dimensions
- Root length ~14.0 mm, sometimes exceeding that of 1st PM

Dimensions	Max. 2 nd PM	Max1st PM
Crown length	8.5 mm	8.5 mm
Root length	~14 mm	14 mm
MD Diameter of Crown	7 mm	7 mm
LL Diameter of crown	9 mm	9 mm

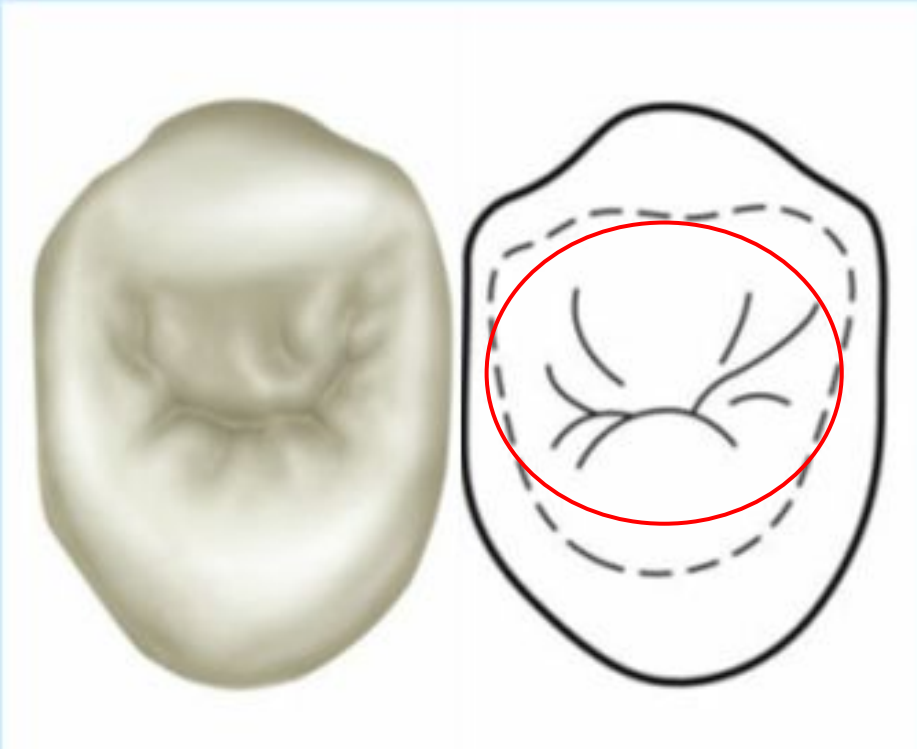


CHRONOLOGY OF THE PERMANENT SUCCEDANEOUS DENTITION

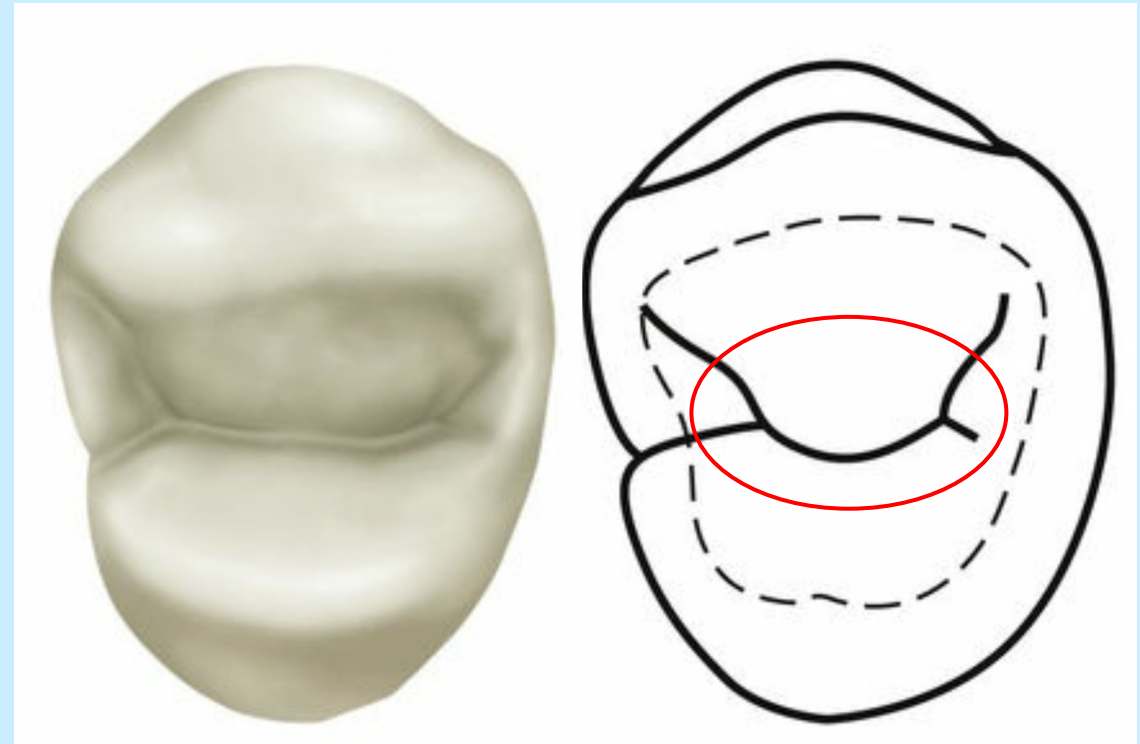
		1st Evidence Of Calcification	Crown Completed (Years)	Eruption (years)	Root Completed (Years)
Central Incisor	Max.	3-4 months	4-5	7-8	10
1st Premolar	Max.	1½–1¾ years	5-6	10-11	12-13
	Mand.	1¼–2 years	5-6	10-12	12-13
2nd Premolar	Max.	2–2¼ years	6-7	10-12	12-14
			4-5 yrs		

Compared to 1st PM, 2nd PM shows:

- More supplemental occlusal anatomy
- Additional secondary ridges and grooves



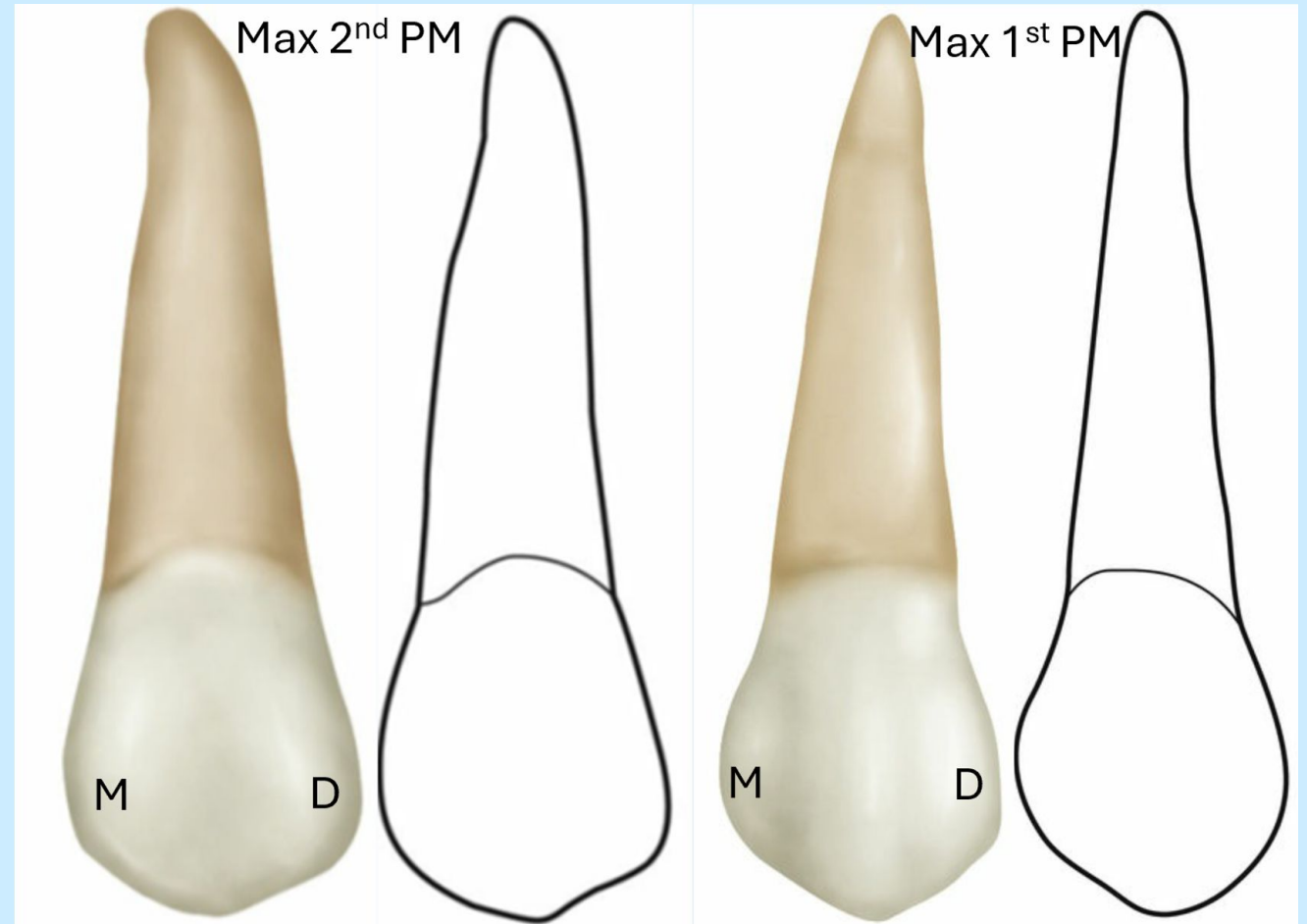
Maxillary left second premolar



Maxillary left First premolar

Buccal Aspect

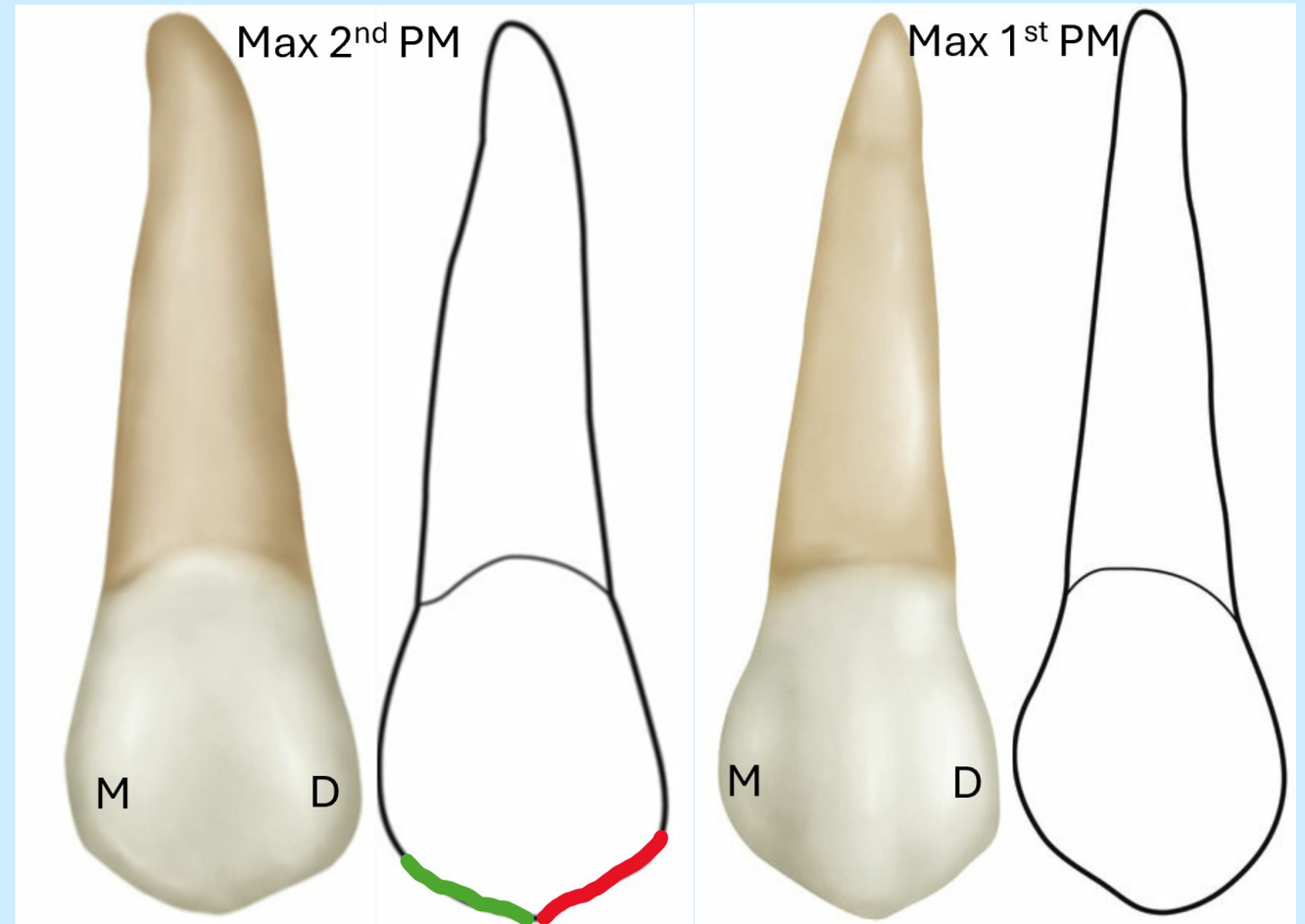
Buccal cusp: smaller
and less pointed than
1st PM



Maxillary left second premolar

Buccal Aspect

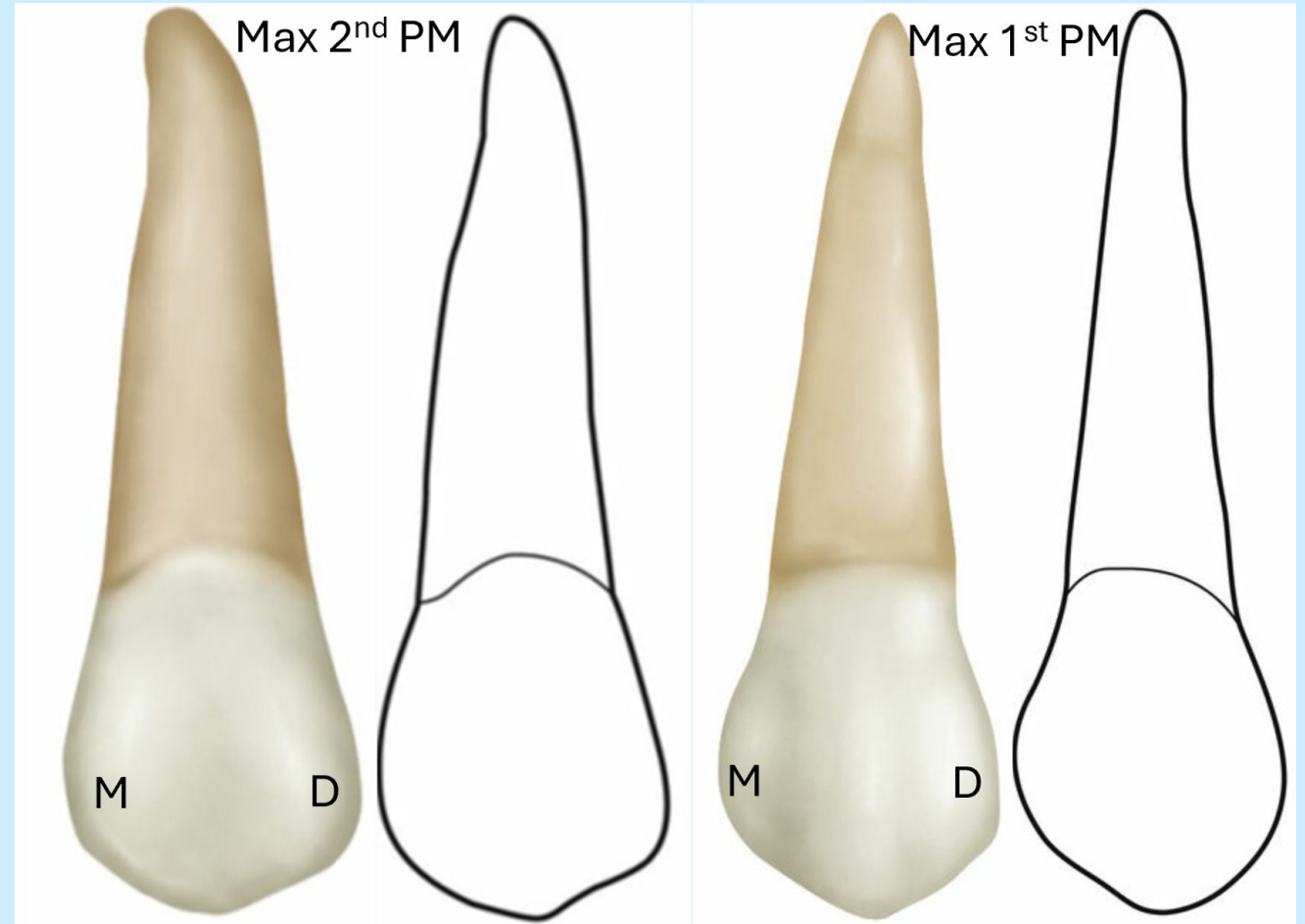
Mesial cuspal slope shorter than distal cuspal slope, that is opposite of max. 1st PM.



Maxillary left second premolar

Buccal Aspect

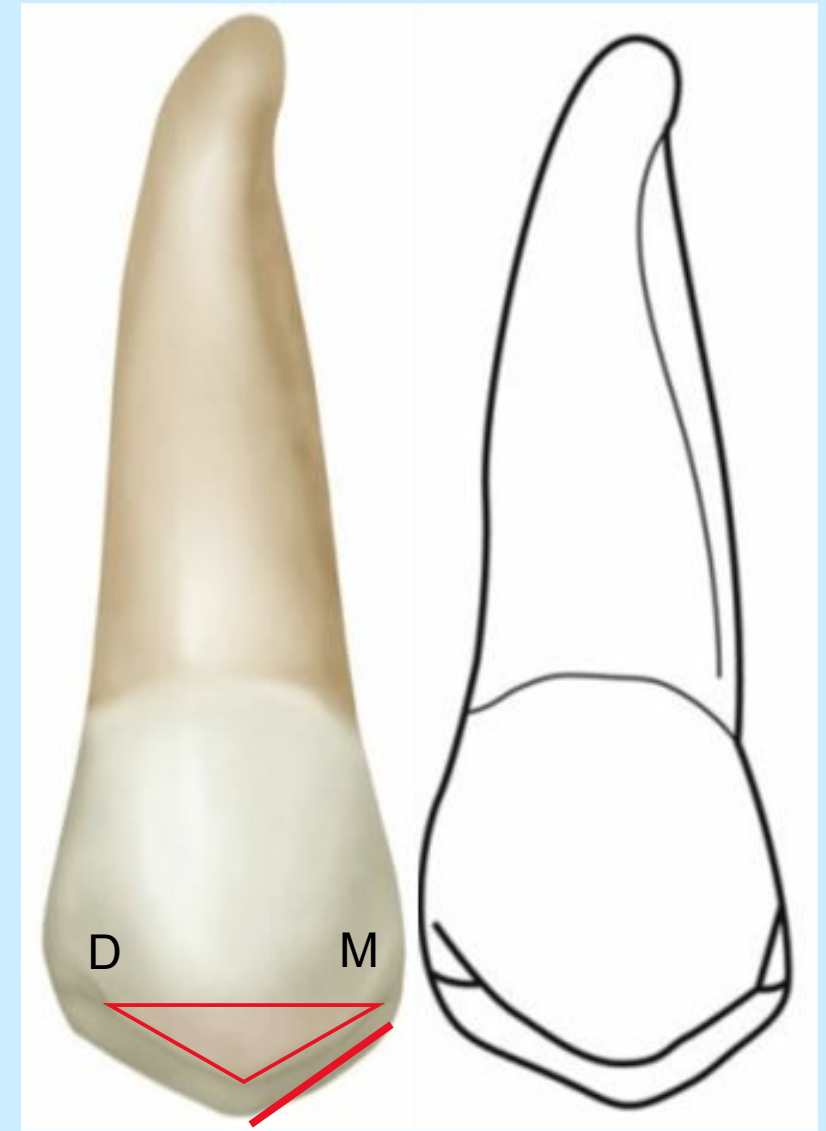
- Buccal ridge less prominent
- Cervical region: crown and root appear thicker (occasionally)



Maxillary left second premolar

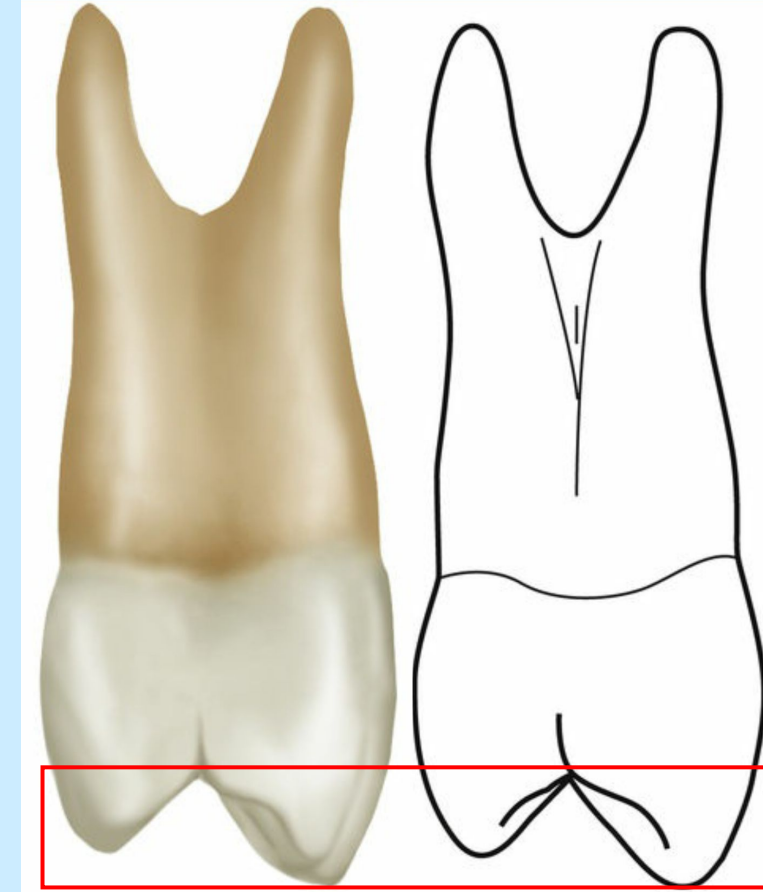
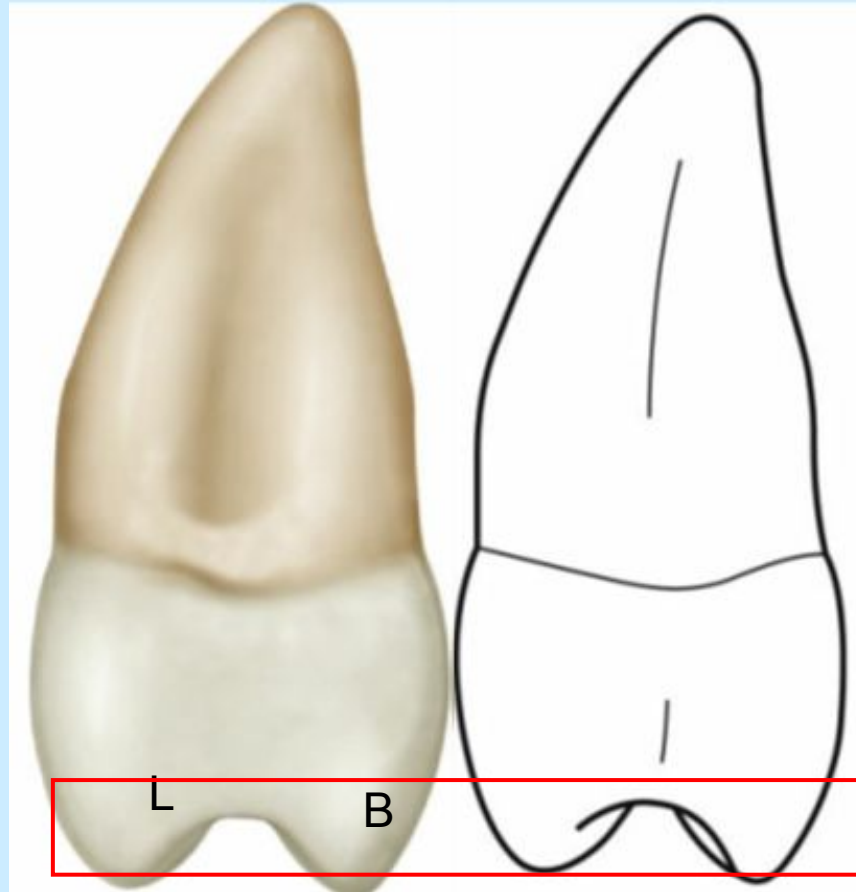
Lingual Aspect

Sometimes, L cusp is longer, which makes the crown longer on the L side



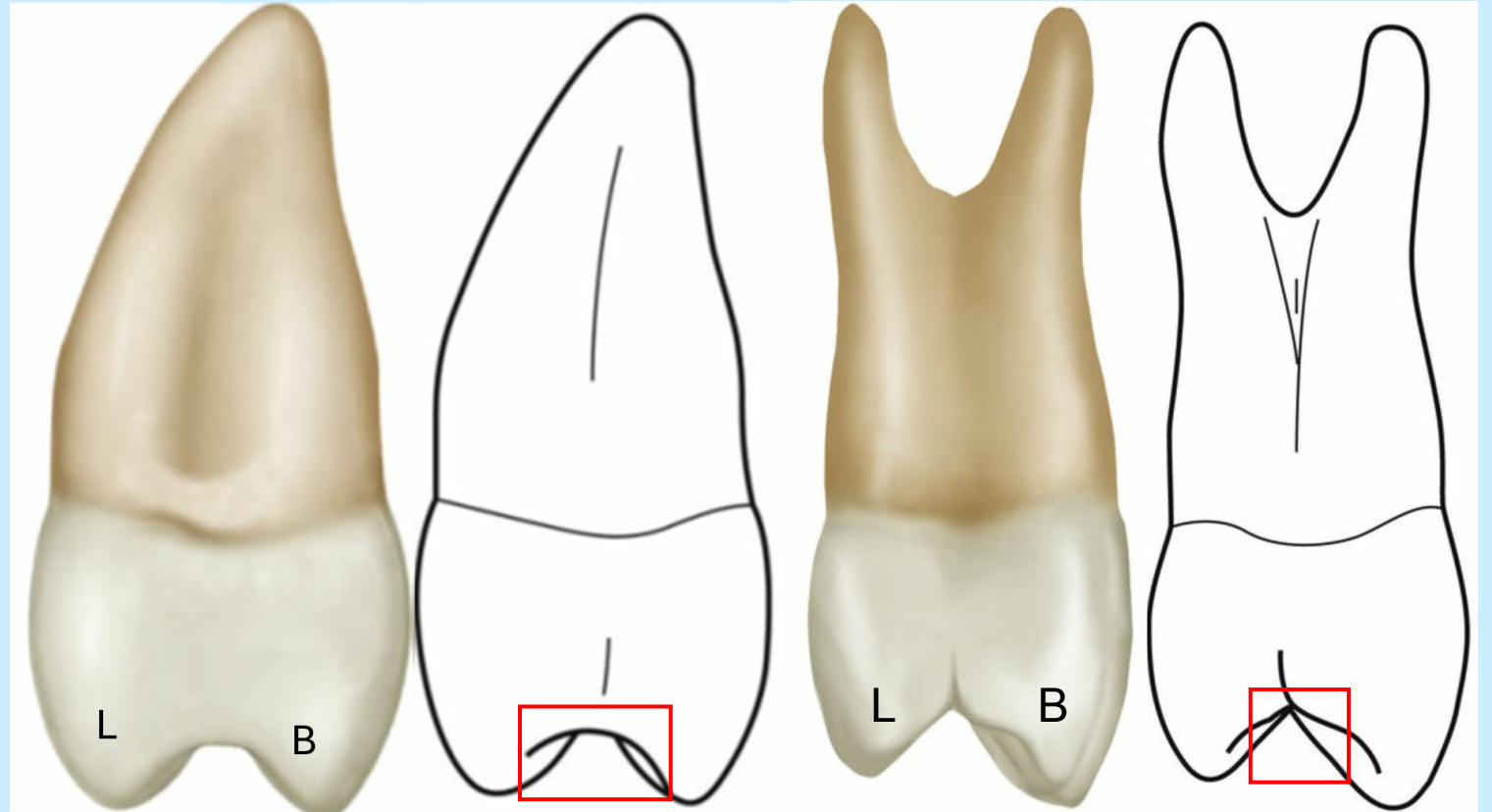
Mesial Aspect

- Cuspal height shorter than max1st premolar;
- buccal & lingual cusps nearly equal in length



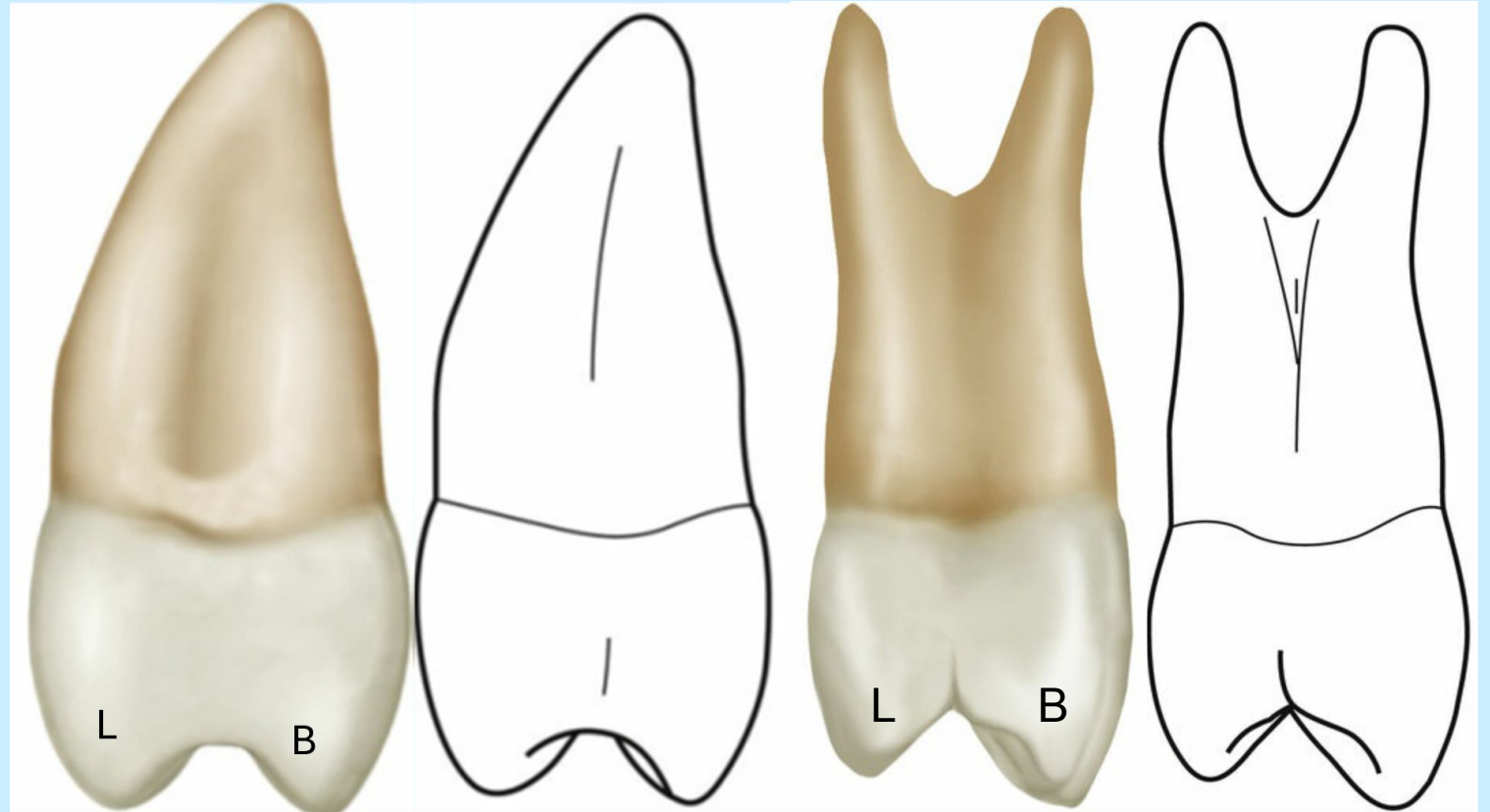
Mesial Aspect

Wider occlusal surface
due to greater BL distance
between cusp tips



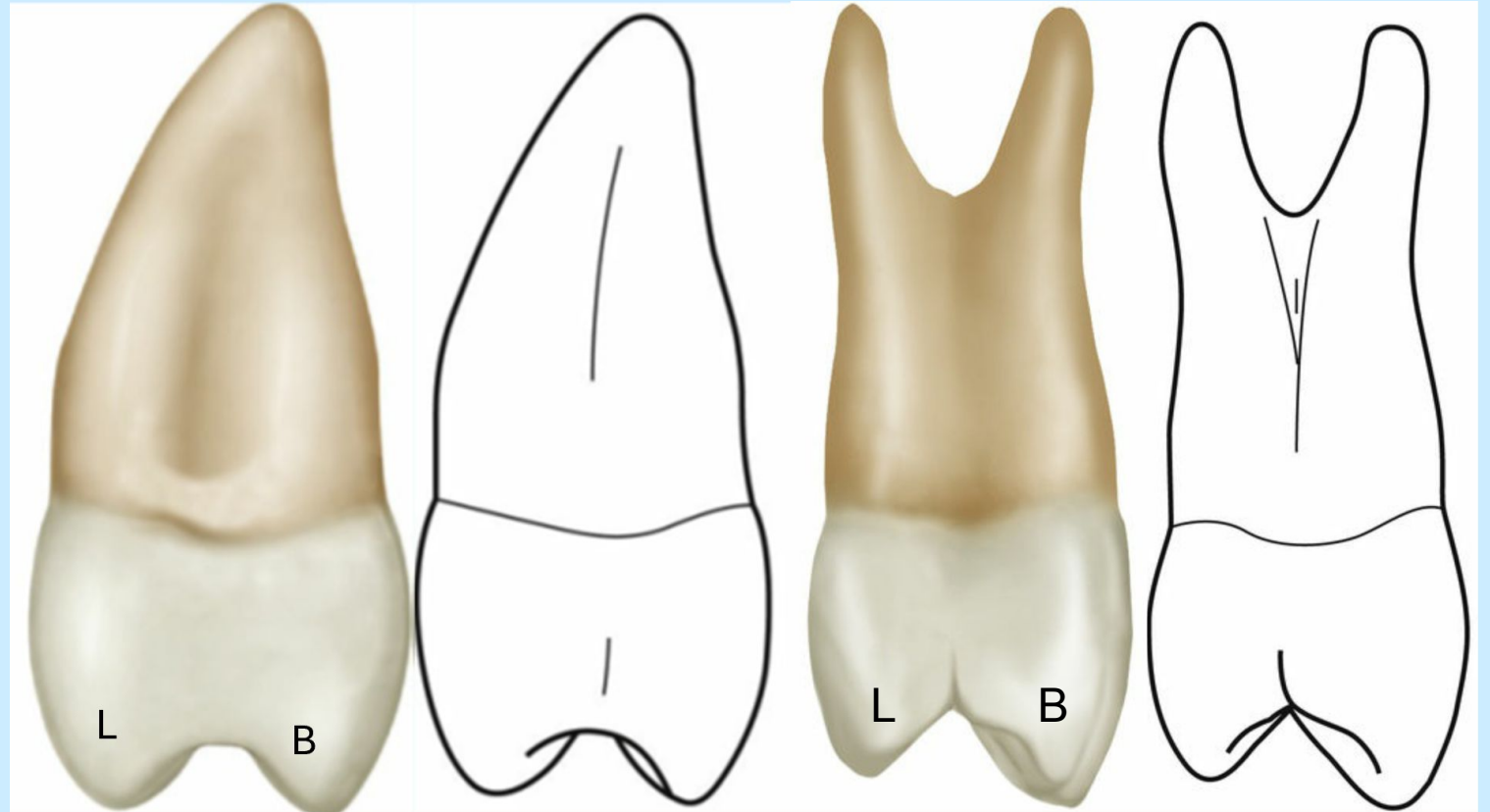
Mesial Aspect

No mesial marginal
developmental
groove present



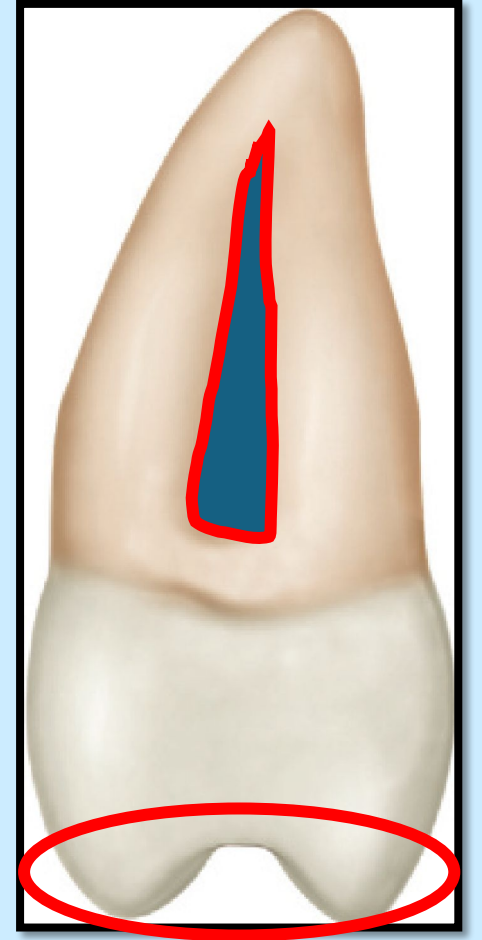
Mesial Aspect

- Crown convex at all points; no developmental depression
- Mesial Root surface shows a shallow developmental groove



DISTAL ASPECT

- The Distal root surface depression is deeper than the Mesial depression on the maxillary 2nd PM.
- This characteristic of the Distal root surface is the opposite of that of the maxillary 1st PM.

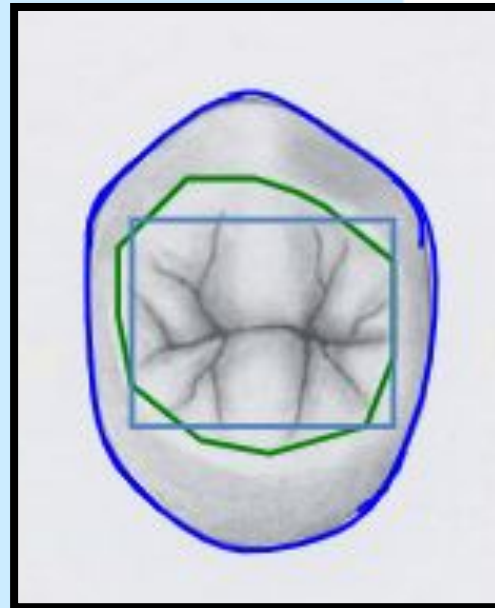
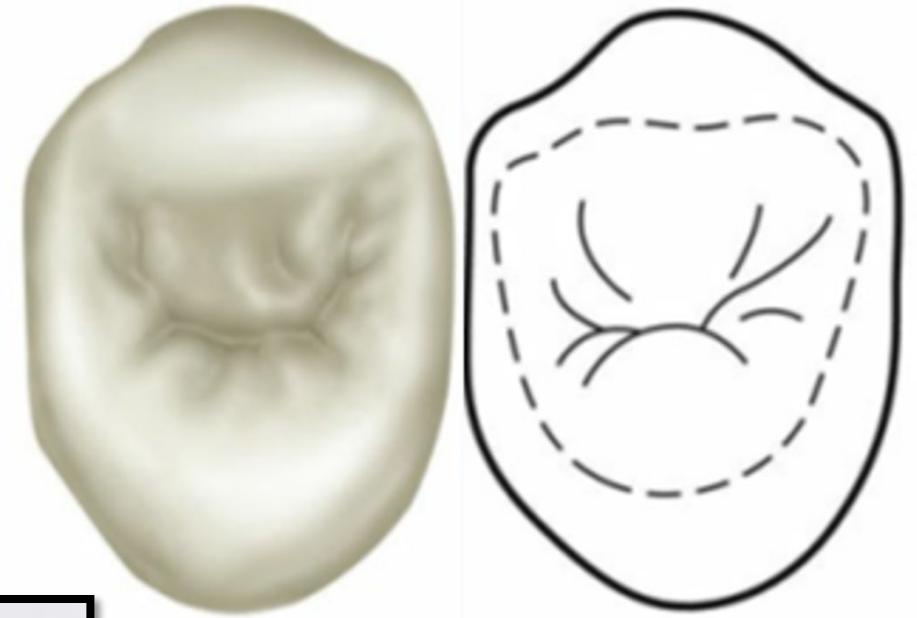


OCCLUSAL ASPECT

Occlusal table Outline:

- **rectangular.**
- “**balanced**” look about the tooth from the occlusal aspect.

Max 2nd PM

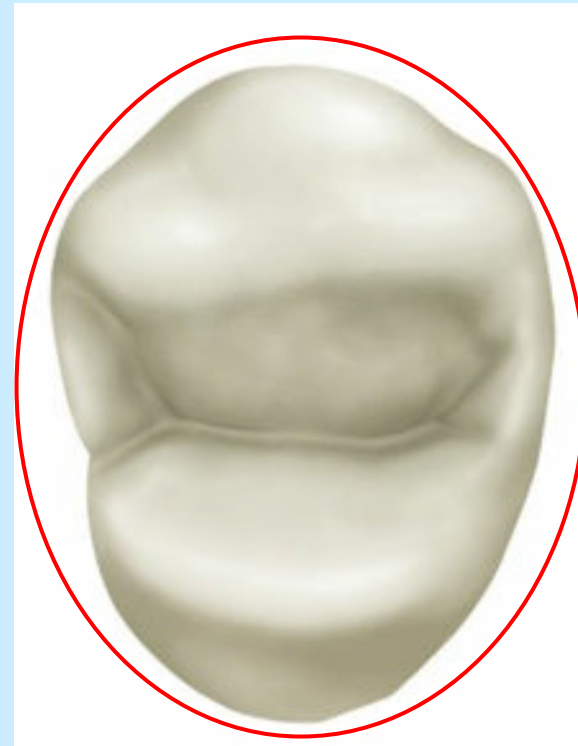


Occlusal Aspect

Outline Less angular, more rounded



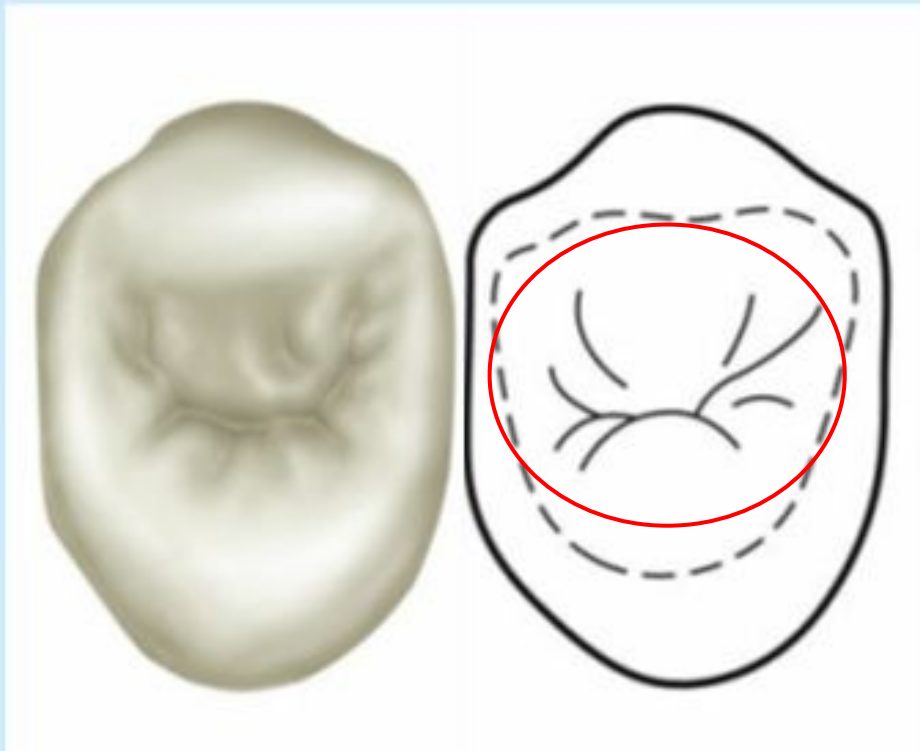
Maxillary left second premolar



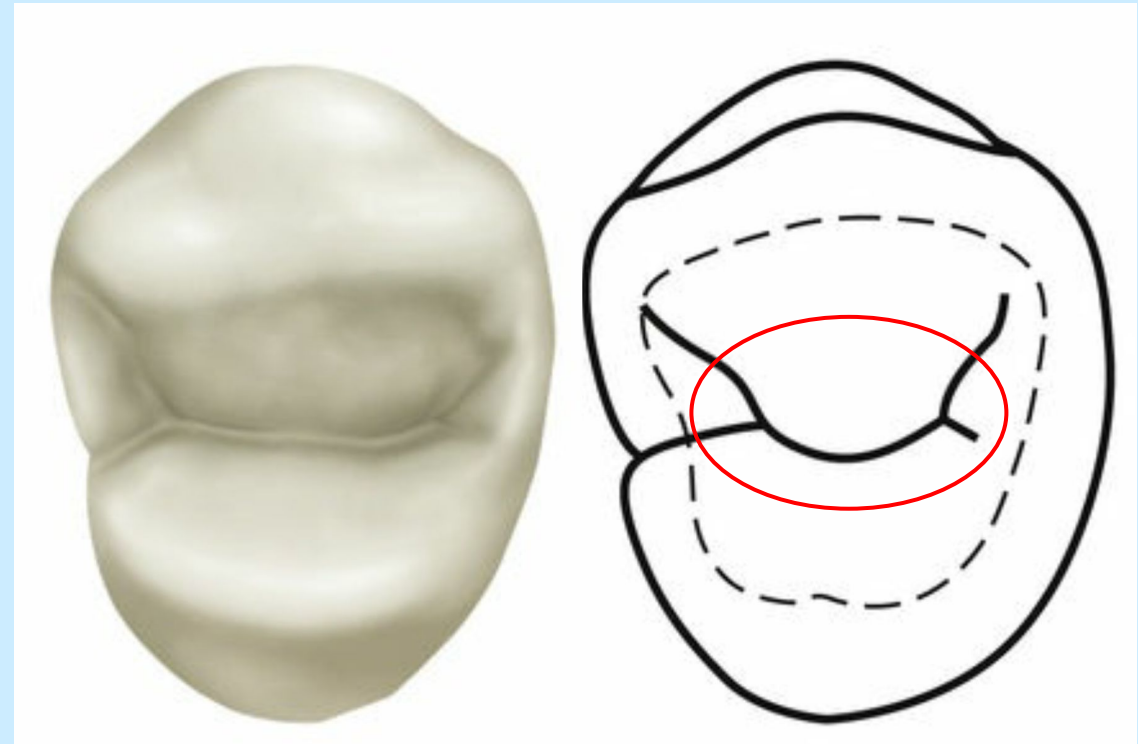
Maxillary left First premolar

Occlusal Aspect

- Occlusal surface irregular with a wrinkled appearance
- Multiple supplemental grooves radiating from central groove



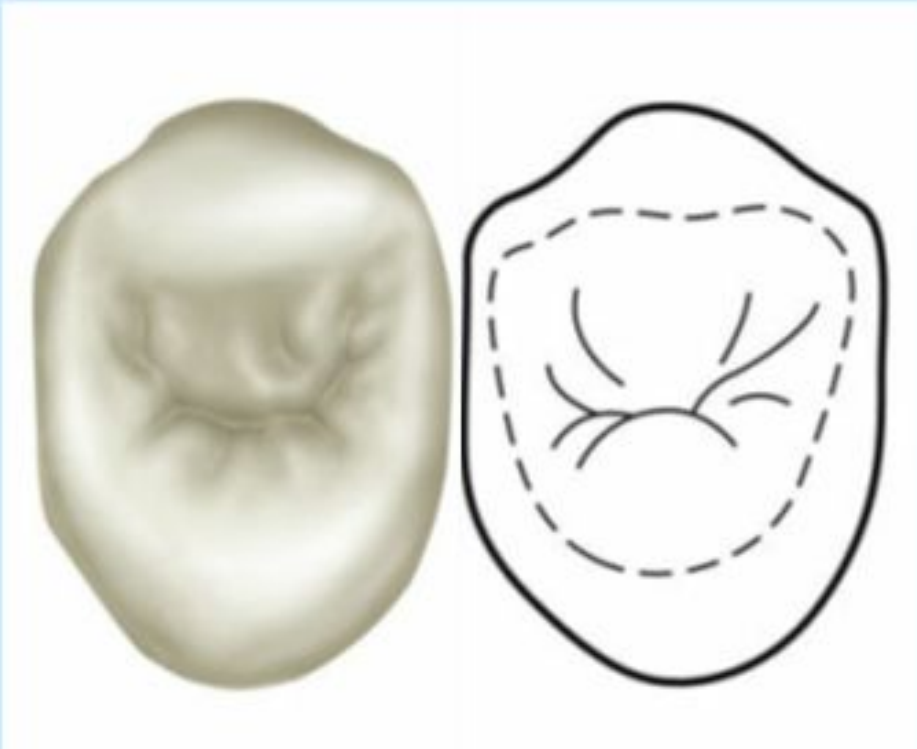
Maxillary left second premolar



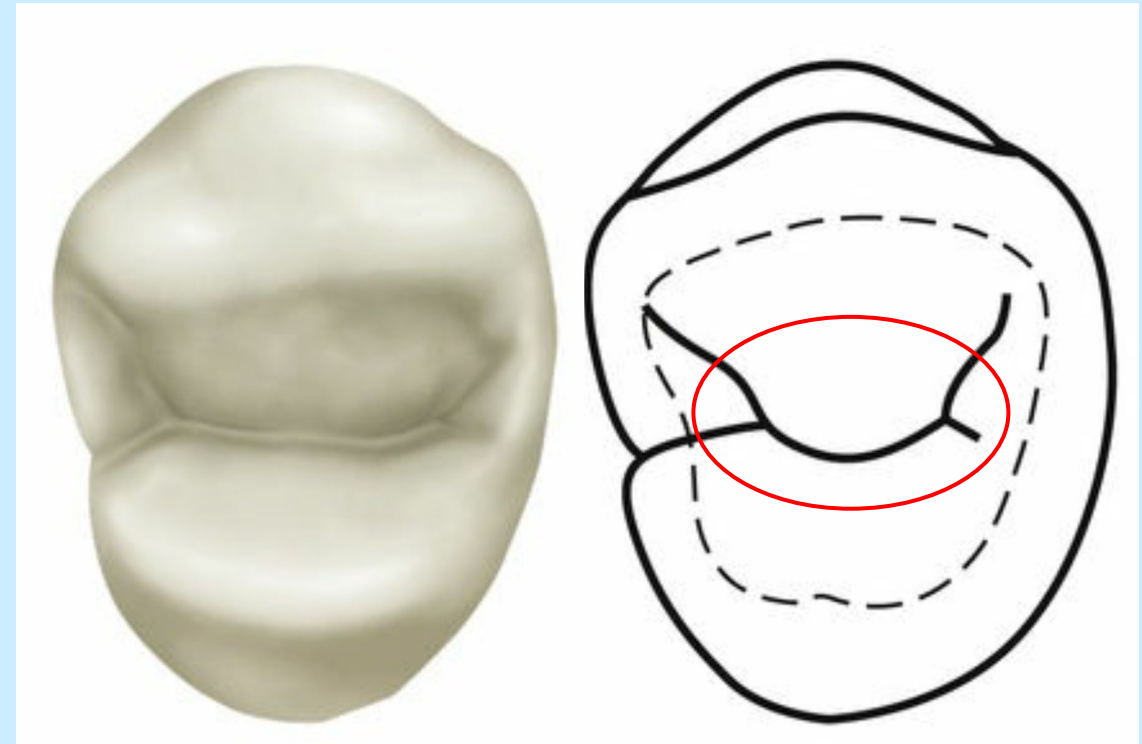
Maxillary left First premolar

Occlusal Aspect

Mesial & distal triangular fossae: irregular, closer together → central groove shorter

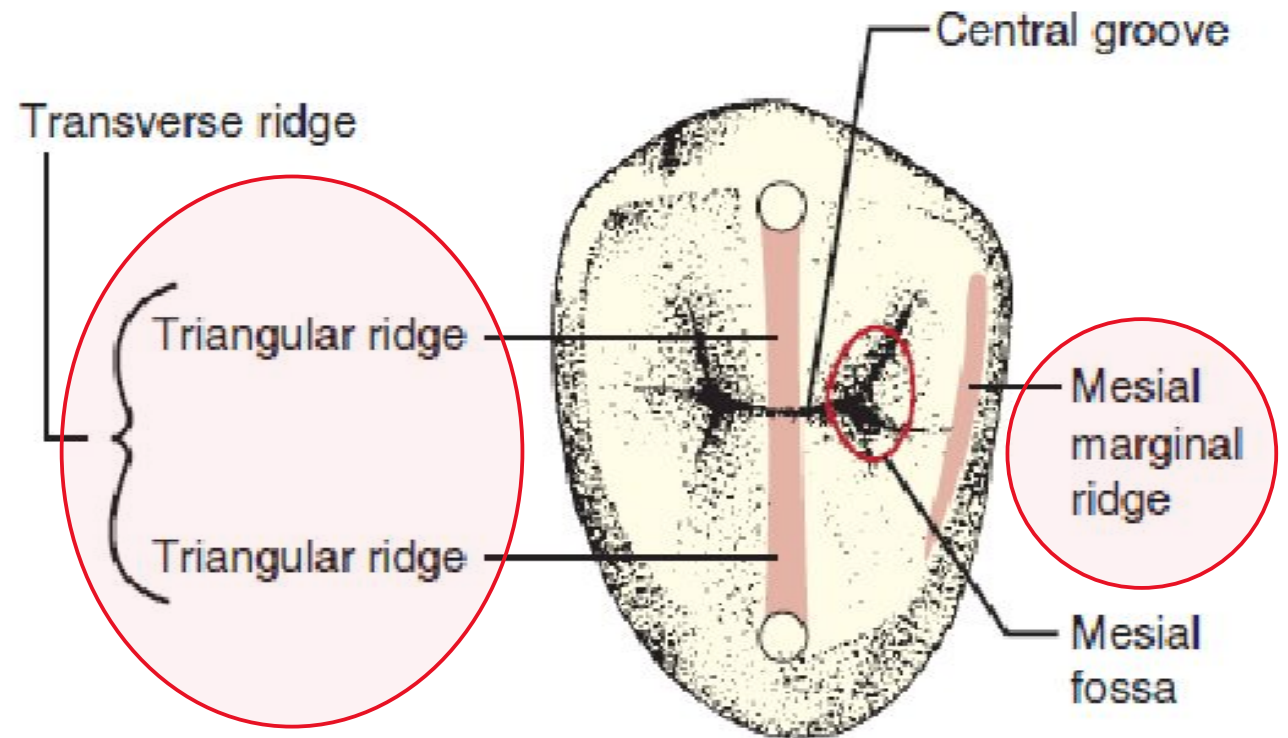


Maxillary left second premolar



Maxillary left First premolar

Triangular ridges of B & L cusps on one side & M Marginal ridge on other side bind the “*Mesial Fossa*”.



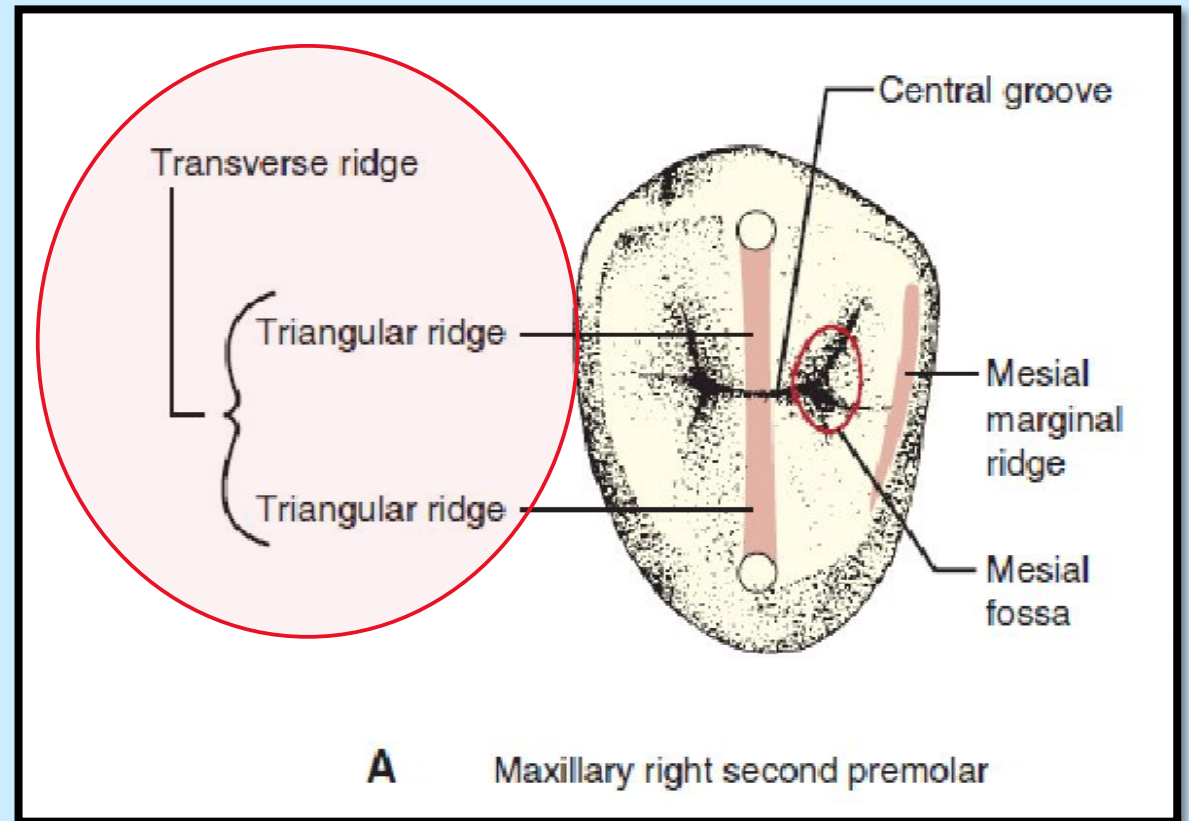
A

Maxillary right second premolar

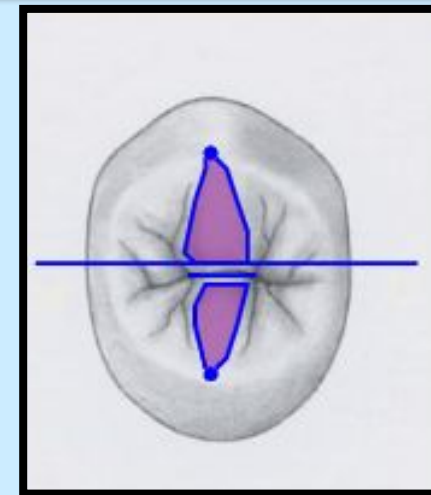
OCCLUSAL ASPECT

Transverse Ridge:

True TR rare as the 2 triangular ridges separated by Central Groove.



B & L Triangular Ridges:
Approx. same length





Any
question

Scenario

While preparing a restoration, a dentist observes that a bicuspid tooth has an **occlusal surface with multiple supplemental grooves**

Question: Which morphological feature is being described?

Options:

- A. Prominent Transverse ridge with adjacent developmental grooves
- B. Wrinkled occlusal surface of maxillary 2nd premolar
- C. Triangular fossae of maxillary 2nd premolar
- D. Mesial marginal groove of maxillary 1st premolar
- E. Triangular ridges of mandibular premolar